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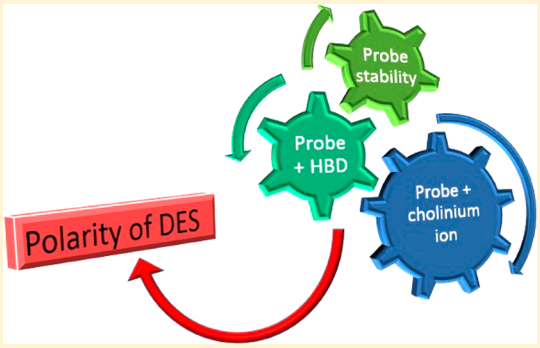
Temperature-Dependent Empirical Parameters for Polarity in Choline Chloride Based Deep Eutectic Solvents

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Supporting Information

ABSTRACT: Deep eutectic solvents have emerged as a promising and economically viable alternative to address our need for environmentally benign media. Efficient utilization of deep eutectic solvents in various fields will require adequate knowledge about their physicochemical properties. In this work, independent polarity scales pertaining to solvent polarizability, dipolarity, acidity and basicity have been determined for some commonly used deep eutectic solvents based on choline chloride. The three eutectic mixtures selected for the study include reline (1:2 mixture of choline chloride and urea), ethaline (1:2 mixture of choline chloride and ethylene glycol), and glyceline (1:2 mixture of choline chloride and glycerol). The variation of the probe response as a function of temperature revealed some interesting details about the polarity of neoteric solvents being investigated. Our observations indicate that while the polarity parameters for ethaline and glyceline are easily determined, the instability of probe molecules in reline poses a potential challenge for the determination of its polarity parameters. Attempts to correlate the empirical polarity parameters with solvatochromic response of Reichardt's dye (E_T^N parameter) also led to surprising observations. Use of alternative probes to measure the polarity in reline indicated that the polarity scales using different probes need to be normalized for DESs. The results emphasize the extent of applicability of currently employed polarity scales to determine the polarity of DES, the need to develop appropriate probe molecules, and the factors limiting our understanding of these extremely interesting class of solvents.



INTRODUCTION

Deep eutectic solvents (DES) are a neoteric class of solvents which have been gaining popularity in the past decade as environment-friendly alternatives to the volatile organic solvents.[1−3] Since the first report on DES by Abbott in 2003,[4] this class of solvents has emerged as a promising alternative medium with many potential applications in organic transformations,[5−7] biocatalysis,[8] biodiesel production,[9−11] extraction of metals,[12,13] CO_2 capture,[14] gas separation/absorption,[15] extraction and purification of bioactive compounds,[16,17] template-assisted synthesis of porous frameworks,[18] electrodeposition of metals,[19] and many more. Deep eutectic solvents are also believed to play a significant role in controlling the physiological functions of organisms.[20] A typical deep eutectic solvent consists of two or more components—with at least one of them being a hydrogen bond acceptor (HBA) and other being a hydrogen bond donor (HBD), resulting in an eutectic mixture. Some commonly used deep eutectic solvents of this class include reline (1:2 mixture of choline chloride + urea), ethaline (1:2 mixture of choline chloride + ethylene glycol), glyceline (1:2 mixture of choline chloride + glycerol) and so on. The resulting eutectic mixtures possess properties similar to ionic liquids (ILs) such as low vapor pressure, high thermal stability, high viscosity and chemical inertness.[3] The similarity in properties often leads to misrepresentation of deep eutectic solvents as another generation of ILs. It should be noted that a deep eutectic solvent may contain charged as well as uncharged moieties, so they are not strictly "ionic" in nature. The nature of solute−solvent interactions in deep eutectic solvents can differ significantly from that observed in ionic liquids. Limited understanding about the solvation processes in DES implies that the choice of the appropriate DES for a given process is primarily a trial-and-error process, guided by intuition and experience. It is critical to understand the solvation properties of deep eutectic solvents by a detailed assessment of relevant solvent properties to exploit their potential uses effectively.

The most common approach to gauge the solute−solvent interactions for a medium is to establish a relative polarity scale using the spectroscopic response of a probe molecule to changes in the cybotactic environment.[21] For molecular solvents as well as ionic liquids, two of most commonly used polarity scales have been Reichardt and Dimroth's normalized E_T scale[22] and the Kamlet−Taft multiparameter scale (α, β, and π^*).[23,24] However, very few studies in the past have focused on the assessment of polarity of deep eutectic solvents. One of the earliest attempts to assess the solvation processes in deep eutectic solvents was reported by Abbott and co-workers for

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eutectic mixtures composed of choline chloride and glycerol.[25] The Kamlet − Taft parameters and $E_T(30)$ parameter were determined for selective compositions of the HBA and HBD component. Since the observed β values were independent of the chloride concentration, the authors concluded that the polarity scale need not necessarily reflect the Lewis basicity of the eutectic solvent.

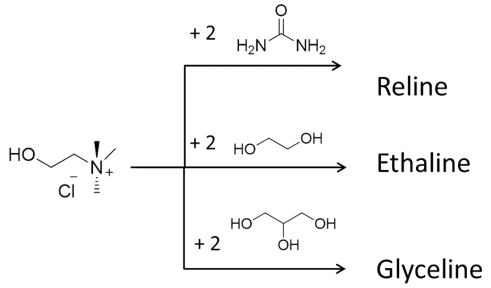
Pandey and co-workers employed the solvatochromic response of several probe molecules to gauge the variation in solvent polarity of four common deep eutectic solvents containing choline chloride as one of the components.[26] The spectroscopic response of the probe molecules was found to be consistent with the previously reported polarity parameters of the eutectics used in the study. The variations in the observed dipolarity led to the conclusion that it is dictated by the choice of the HBD component. Pandey et al. also measured the Kamlet−Taft parameters and Reichardt's E_T^N scale for pure DES (reline, ethaline and glyceline) as a function of temperature and concentration of water.[27] Recently, the study of solvatochromic parameters have been extended to deep eutectic solvents composed of ammonium salts and carboxylic acids.[28]

An important factor affecting the solvation properties and hence the measured polarity parameters, is the presence of molecular cosolvents as impurity or through deliberate addition. For example, given the hygroscopic nature of many deep eutectic solvents, it is but natural to expect traces of water as an impurity in the DES sample. This has prompted chemists to investigate the variation in polarity due to the presence of additional cosolvents. In the study by Pandey et al., the Kamlet−Taft parameters were found to increase linearly with increasing mole fraction of water in aqueous DES mixtures.[27] Interestingly, when pyrene was used as a polarity probe, the response was higher for the DES + water system than either of the pure components (DES or water)—a phenomenon termed as "hyperpolarity". The experimentally determined pyrene signals deviate from ideal behavior for all three DES + water systems, glyceline−water mixtures being the most deviant and reline-water systems being the least. Such observations led to the conclusion that small amounts of molecular cosolvents can drastically alter the solvation capabilities of the deep eutectic solvents. In another investigation on tetratethylene-modified DES, it was found that addition of small amounts of cosolvent significantly altered the polarity of the DES, leading to the proposal that such systems are indeed "hybrid" solvent systems.[29] Interesting variations in polarity were also observed for mixtures of reline and glyceline with molecular solvents like 2,2,2-trifluoroethanol (TFE) and hexamethylphosphoramide (HMPA).[30] Such polarity studies based on spectroscopic behavior of probes have also been carried out for mixtures of common DES like reline, ethaline and glyceline with DMSO, which indicated that the addition of DMSO led to an increase in the hydrogen bond accepting ability and a simultaneous decrease in the hydrogen bond donating ability.[31]

It should be noted that this approach has some potential limitations. The reader is referred to a recent article wherein the advantages and limitations of extending the conventional polarity scales for molecular liquids to ionic liquids have been critically analyzed and appropriate modifications have been suggested.[32] The most important factor to be considered is that polarity scales based on solvatochromic behavior of probes are not universal in nature i.e. they may be specific to the particular probe employed for the study. Thus, a given polarity parameter determined using different probe molecules for different solvent systems cannot be compared. In ionic liquids and deep eutectic solvents, the presence of Coulombic forces induces variation in the nature of solute−solvent interactions, depending on whether the solute is charged or uncharged. It was observed by Catalan and co-workers that any suitable probe designed for measuring a particular aspect of solvent polarity will also experience specific solvation to a greater or lesser extent and thus requires the use of a homomorph of the probe to offset contributions of specific effects.[33] On the basis of this approach, Catalan and group developed independent polarity scales for solvent polarizability (SP),[34] dipolarity (SdP),[35,36] acidity (SA),[37] and basicity (SB).[38] After establishing the generalized scale for a range of molecular solvents covering a broad polarity range,[33] the validity of the scale was tested with respect to ionic liquids.[39]

In the present report, Catalàn's generalized scale of polarity is applied to three deep eutectic solvents and the results are compared to previously reported polarity parameters, in order to understand the solvation process in deep eutectic solvents. The three DES (Scheme 1) employed in the present study—

Scheme 1. General Representation of Reline, Ethaline, and Glyceline



reline, ethaline, and glyceline—have been chosen since they are most widely used for numerous applications. The effect of temperature on the solvatochromic response of the probes reveals some very interesting details about the polarity of the eutectic mixtures studied. Finally, the observed polarity parameters are discussed vis-a-vis the physicochemical properties of the DES in order to outline the potential areas for future investigation.

MATERIALS AND METHODS

Choline chloride, urea, ethylene glycol, and glycerol employed for synthesis of DES were of highest purity available. Deep eutectic solvents were prepared by mixing choline chloride with hydrogen bond donors in appropriate mole ratios followed by stirring and heating (~90 °C) until a homogeneous, colorless liquid has been formed.

Anthracene and 5-nitroindoline used for spectroscopic determination of polarity were of the highest purity available. The molecular probes used—1-methyl-5-nitroindoline (**MNI**),[38] 3,6-diethyltetrazine (**DETZ**),[40] 2-*N*,*N*-dimethylamino-7-nitrofluorene (**DMANF**),[35,36] 2-fluoro-7-nitrofluorene (**FNF**),[41] 4-*tert*-butyl-2-(dicyanomethylene)-5-[4-(diethylamino)benzylidene]-Δ^3-thiazoline (**Th**),[42] and 2-[4-(*N*,*N*-dimethylamino)benzylidene]malononitrile (**BMN**)[43]—have been synthesized as per reported procedures and subject to chromatographic isolation to ensure high purity. Stock solutions of all probes were prepared by dissolving them in

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ethanol in amber glass vials (ethanol was found to be the most convenient medium for introducing the dyes in the DES, as per previously reported procedure[27]). An appropriate amount of the probe solution from the stock was transferred to the quartz cuvette containing DES. After uniform dissolution of probe solution in DES, complete removal of ethanol was ensured by application of high vacuum for more than 30 min. The solution obtained was used for spectroscopic measurements. Temperature dependent spectroscopic measurements were carried out by varying temperature from 298 to 323 K (interval of 5 K) with UV−vis spectrophotometer equipped with temperature controller. All spectroscopic measurements were performed at least in triplicate starting from sample preparation, and the results were averaged.

RESULTS AND DISCUSSION

The four independent solvent polarity scales (SP, SdP, SA, and SB) were initially determined for the three commonly used deep eutectic solvents—namely reline, ethaline, and glyceline between 298 and 323 K using the standard procedure reported in literature (see Supporting Information for details). While measuring the temperature-dependent SdP values for reline, it was observed that the absorbance intensity of the probe **DMANF** decreased drastically at higher temperatures, indicating that the probe may be unstable in reline. No decrease in absorption intensity was observed with increasing temperatures for ethaline and glyceline. In the past, it has been reported that the solvents with high acidity preclude the use of **DMANF** as a probe due to its tendency to undergo protonation in such media. In order to overcome the limitation, SdP is determined using its correlation with the dipolar reaction field function (Δf), which is a function of static dielectric permittivity (ε) and refractive index (n).[33] This approach was unfeasible in the present case since the values of dielectric permittivity are not yet reported for reline. Another approach involves estimation of a linear correlation between the absorption frequencies of **DMANF** and its homologue **FNF**, which allowed the determination of the absorption frequency of **DMANF** and SdP values, in alcoholic solvents.[36] A plot of ν_{DMANF} versus ν_{FNF} for ethaline and glyceline at different temperatures (Figure 1) yielded a linear relation eq 1

$$\nu_{DMANF} = 1.869(\pm 0.131)\nu_{FNF} - (32.327(\pm 3.850) \times 10^3)$$
$$(n = 12,\ r = 0.9764,\ \text{and SD} = 5.827) \quad (1)$$

It must be noted that the SdP parameter for reline should be regarded as a secondary value since it has been derived from correlations between analogues and not by direct experimental measurements. The estimation of SA and SPP parameters also relies on ν_{DMANF} and hence, some caution may be exercised while discussing the same. The major drawback of this approach is that it assumes that the solvatochromic response of the **DMANF** probe in reline is very similar to that observed in ethaline and glyceline. Such an assumption is justified in the earlier correlation which was applied only to alcohols,[36] but it may not be entirely justified for correlating the probe behavior in reline (containing a diamine as HBD component) with that observed in ethaline and glyceline (containing polyols as HBD components). Any further discussion on the resulting polarity parameters of reline will be seriously limited and its implications will be observed in the following discourse.

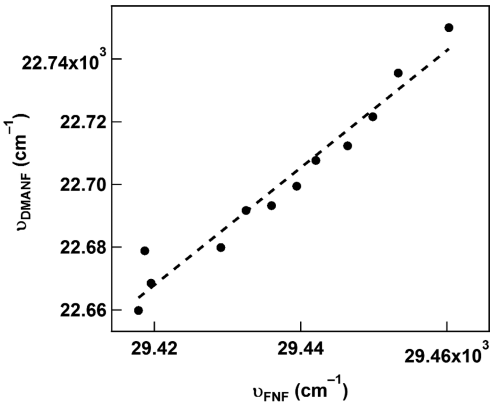


Figure 1. Correlation between ν_{DMANF} and ν_{FNF} at different temperatures for ethaline and glyceline (the dotted line is the linear fit).

The SdP values for reline were then estimated using this linear correlation and the resulting polarity parameters for reline at 298 K are listed in Table 1 along with the polarity values for ethaline and glyceline.

Table 1. Empirical Solvent Parameters for Reline, Ethaline, and Glyceline at 298 K

DES	SP	SdP	SA	SB
reline	0.877_2	0.763_9[a]	0.665_9	0.358_2
ethaline	0.815_5	0.968_5	0.550_5	0.378_1
glyceline	0.851_1	0.872_2	0.585_2	0.425_7

[a]Determined using linear correlation between ν_{DMANF} and ν_{FNF}

The validity of the scales was verified by comparing the SP values with the corresponding polarizability values obtained by using the Lorentz−Lorenz equation. The refractive index values required for applying the Lorentz−Lorenz equation were taken from previously reported literature for deep eutectic solvents[44−46] and the resulting correlation is shown in Figure 2. (See Supporting Information for the refractive index values.)

Strong linear correlation is observed between the Lorentz−Lorenz polarizability function and SP values measured at different temperatures for reline (n = 6, r = 0.9722 and SD = 0.0003) and ethaline (n = 4, r = 0.9969 and SD = 0.0004). The lower number of data points used for comparing the SP scale for ethaline is the result of paucity of adequate refractive index data. The analysis could not be extended to glyceline, again due to lack of data on temperature dependent refractive index. Nevertheless, the good correlation between the SP scale and the Lorentz polarizability for reline and ethaline show the consistency of this scale for DES.

Experimentally, SPP is determined as a function of the difference in the absorption frequencies of **DMANF** and **FNF** probes, i.e., $SPP_{exp} = ((\Delta\overline{\upsilon}_{solvent}) - 4692)/2119$, where $\Delta\overline{\upsilon}_{solvent} = \overline{\upsilon}_{DMANF} - \overline{\upsilon}_{FNF}$. The calculated values of SPP are derived from a linear correlation between SP and SdP scales, which was shown to be valid for molecular solvents as well as ionic liquids−given as SPP_{calc} = (0.49 ± 0.03) SP + (0.38 ± 0.01) SdP + (0.21 ± 0.02). The resulting correlation is depicted in Figure 3 for ethaline (SPP_{exp} = −0.2 (±0.1) + 1.22 (±0.15) SPP_{calc}, n = 6, r = 0.970, SD = 0.001) and glyceline (SPP_{exp} = −0.026 (±0.09) + 1.22 (±0.10) SPP_{calc}, n = 6, r = 0.981, SD = 0.025), yielding a slope close to unity in both the cases. The observations verify that the contribution of the SP and SdP

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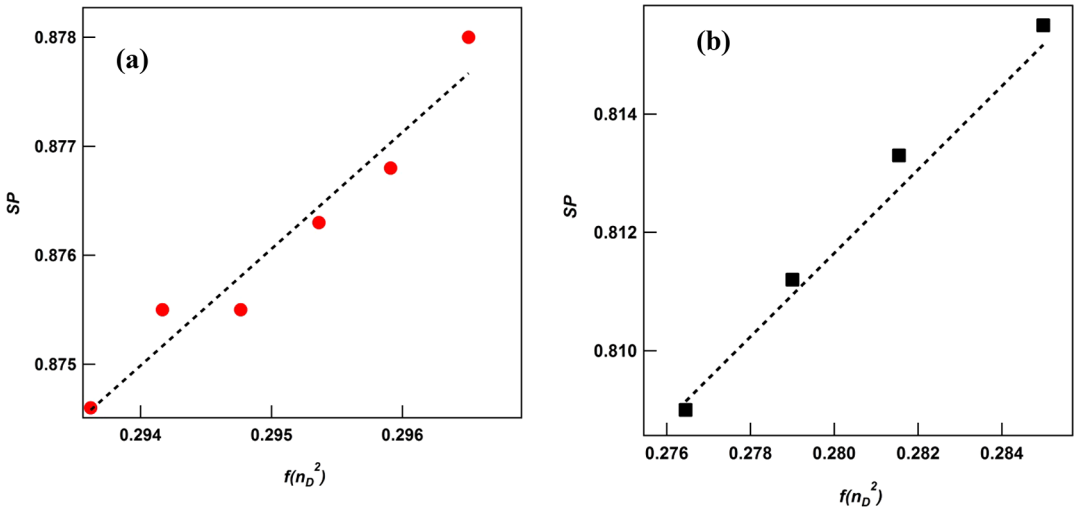


Figure 2. Correlation between temperature-dependent solvent polarizability (SP) scale and the Lorentz−Lorenz function for (a) reline and (b) ethaline.

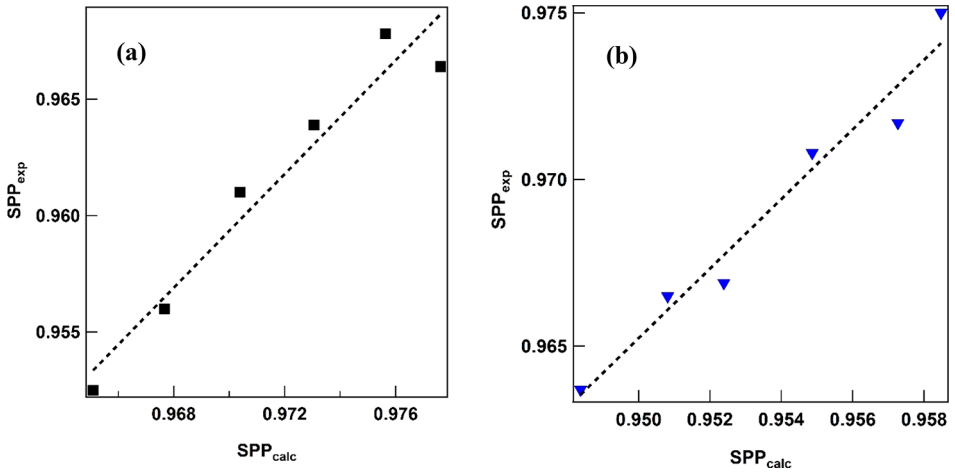


Figure 3. Correlation between experimental and calculated solvent dipolarity and polarizability (SPP) scale for (a) ethaline and (b) glyceline.

parameters toward the SPP parameter in ethaline and glyceline is the same as that observed for molecular solvents and ionic liquids—thus verifying the universal nature of the correlation between SPP values and SP/SdP values. Since the SdP values for reline have not been determined by direct measurements, we decided to exclude them from this comparison.

In order to evaluate the validity of the SdP parameter obtained for reline, the estimated SdP values for the three DES in the present work were compared qualitatively with the values reported in literature using different solvent dipolarity probes. In an earlier report, Pandey and co-workers had used the solvatochromic response of different fluorescent probe molecules in addition to Reichardt's dye (30 and/or 33) to assess the dipolarity of four DES, which included reline, ethaline and maline.[26] It was observed that almost all the probes, with the exception of pyrene and its homologue pyrene-1-carboxaldehyde, indicated that ethaline and glyceline were more dipolar than reline. Most notably, the fluorescent response of 1-anilino-8-naphthalenesulfonate (ANS), *p*-toluidinyl-6-naphthalenesulfonate (TNS), and Nile Red indicate that the dipolarity of the solvents follow the order: ethaline > glyceline > reline, which is in close agreement with the SdP trend reported in Table 1. Other probes, such as 6-propionyl-2-(dimethylaminonaphthalene) (PRODAN) and coumarin-153 also indicate that reline has the least dipolar among the three solvents with ethaline and glyceline having similar dipolarity (PRODAN reports marginally higher dipolarity for glyceline than ethaline while coumarin-153 indicates similar response for the two). Given the fact that the contribution of probe-specific solvation cannot be ruled out for any measurement of polarity, it is indeed remarkable that the trend in literature values agree with the SdP values reported for the three solvents, thus qualitatively verifying the SdP estimate for reline.

The thermosolvatochromic response of all the four polarity scales in reline, ethaline, and glyceline was evaluated (Figure 4) and a good to satisfactory linear correlation with temperature from 298 to 323 K was observed for all of them. This should allow estimation of polarity values at intermediate temperatures (through interpolation) and at lower or higher temperatures (through extrapolation, up to a certain limit). There are some very interesting differences in the thermosolvatochromic behavior of the three solvents. The solvent polarizability (SP) for all the three solvents shows marginal dependence on temperature as indicated by the low magnitude of the slopes (Table 2). The solvent dipolarity (SdP) for ethaline and glyceline shows a marginal negative thermosolvatochromic

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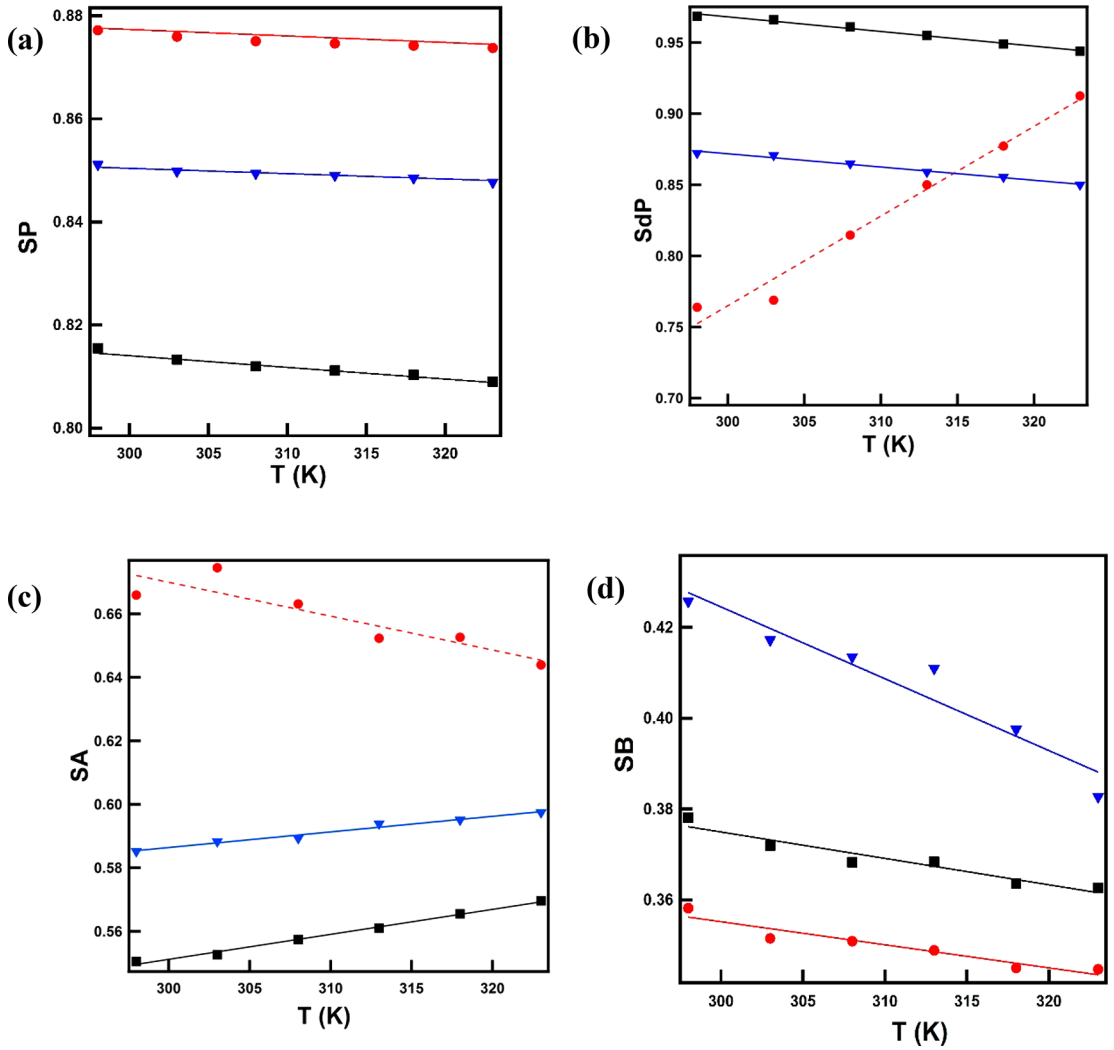


Figure 4. Temperature-dependent response of SP, SdP, SA, and SB parameters for reline (red), ethaline (black), and glyceline (blue). The dotted lines for the SdP and SA data of reline are used to emphasize the fact that the parameters are not empirical but are indirectly determined by a correation.

Table 2. Temperature Dependence of Solvent Polarity Parameters for DES from 298 to 323 K in Terms of the Slopes Obtained from Least Squares Fitting

	temperature dependent response			
DES	SP ($\times 10^{-4}$ K^{-1})	SdP ($\times 10^{-3}$ K^{-1})	SA ($\times 10^{-3}$ K^{-1})	SB ($\times 10^{-4}$ K^{-1})
reline	−1.29 (±0.17)	6.31 (±0.46)	−1.07 (±0.26)	−5.08 (±0.81)
ethaline	−2.42 (±0.22)	−1.03 (±0.06)	0.78 (±0.04)	−5.83 (±0.84)
glyceline	−1.22 (±0.13)	−0.93 (±0.06)	0.49 (±0.04)	−15.81 (±2.7)

[a]Determined using least-squares fitting of the traces in Figure 4

shift. However, the SdP values of reline increase drastically with an increase in temperature. The contradictory behavior of reline vis-à-vis that observed for ethaline and glyceline also extends to the solvent acidity—reline has the highest SA values of the three solvents and the values show inverse relation with temperature while ethaline and glyceline show positive thermosolvatocromism but to a smaller extent. For the solvent basicity, the SB values for ethaline are the highest and show the largest dependence on temperature. The temperature-dependent behavior of ethaline and glyceline mimics that observed for ionic liquids [BMIM][BF_4] and [BMIM][PF_6].[39]

It must be noted that the polarity of solvents is expected to decrease with an increase in temperature and almost all the molecular solvents confirm to the norm, which led to the generalization of "thermochromic red shift".[47] However, a notable exception to this rule was reported for the variation in Kamlet Taft polarity parameters of pyrrolidinium and pyridinium ionic liquids, which showed positive thermosolvatochromism in some cases.[48] The opposing behavior of different ionic liquids was justified on the basis of difference in the extent of stabilization of the ground state and excited state of the dye molecules by solvation. For ionic liquids which showed an increase in polarity with temperature, it was concluded that the excited state was stabilized to a greater extent in such solvents, which was adversely affected by the increase in temperature. In case of mixtures such as DES, it is

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also possible that the polarity probe interacts preferentially with one of the components. Increase in temperature weakens the extent of association between the HBD and HBA components and hence, facilitating better interaction with the polarity probe. However, more detailed analysis of the polarity trends in a large number of DES needs to be undertaken before stating the actual forces responsible for the positive shift in polarity with temperature.

The difference in the thermosolvatochromic variation of SdP and SA parameters in reline as compared to ethaline and glyceline may be traced to the differential effect of urea versus ethylene glycol (or glycerol) as hydrogen bond donating (HBD) components of the solvent, since choline chloride is common in all three of them. In urea-based deep eutectic solvents, the $-NH_2$ functional groups are the main hydrogen bond donor site while for ethaline and glyceline, it is the alcoholic −OH functional groups. Hence a higher hydrogen bond donating capability in the latter class of DES and the observation of higher values of SA is justified. An increase in temperature may potentially affect the interaction of the −OH and $-NH_2$ functionalities in different manners, thus inducing responses which are opposite in nature. The dissimilarity is more obvious for solvent parameters which are more dependent on the identity of the HBD—namely the SdP and SA parameters. Alternatively, the difference may arise from the fact that SdP and SA of reline have been determined by a correlation of the **DMANF** response in the other two DES (Figure 1). The anomalous deviation may also arise if the assumption that **DMANF** experiences structurally similar solvation in the three DES is not valid. In order to address this concern, a more generally applicable probe molecule may be used for determination of SdP and by extension, SA in DESs.

A comparison of the observed polarity parameters for the three DES with those reported for the commonly used solvents (such as water, DMSO, DMF, etc.) and two ionic liquids for which these parameters have been reported is listed in Table 3. In order to facilitate comparison with the polarity parameters of molecular solvents which are reported at 293 K, the polarity values for the three DES at 293 K were estimated by extrapolation of the experimental data and use of linear fits. The values of SP, SdP, SA, and SB for reline, ethaline, and glyceline thus obtained should be within reasonable accuracy to allow a qualitative comparison with the polarity of conventional solvents. The table also lists the polarity values for the molecular solvents which have comparable polarity, which leads to some interesting observations. For example, the SP values of all three DES are higher than many molecular solvents and even ionic liquids. The SP of reline is comparable to that observed for aromatic solvents like bromobenzene (SP = 0.875) or dibenzyl ether (SP = 0.877). However, the solvent acidity and basicity parameters are much higher for reline than either of the two molecular solvents. Ethaline has SP value similar to formamide (SP = 0.814) while glyceline SP values are closer to the values observed for acetophenone (SP = 0.848), 2-phenyl ethanol (SP = 0.849) or benzonitrile (SP = 0.851). With the exception of ethaline, the SdP values for the other two DES are lower than most commonly used polar solvents. The overall polarity of the deep eutectic solvents is closer to that observed for formamide, 1,2-ethanediol and 1,2,3-propanetriol, though not necessarily in that order. Formamide shows the most similarity with ethaline and glyceline when all the polarity parameters are taken in account while 1,2,3-propanetriol corresponds the most with reline. The similarity of polarity parameters between ethaline and 1,2-ethanediol and that between glyceline and 1,2,3-propanetriol indicates that the HBD component of the DES has a significant contribution to the observed polarity of the DES, except for SP parameter which is much higher. However, it is important to test the polarity of many more DES before a general conclusion can be drawn in this regard. This observation may also help in justifying the aforementioned argument that a few of the polarity probes which interact preferentially with the HBD component of the DES, may undergo positive thermosolvatochromism. The comparison brings out the unique combination of polarity parameters in DES which were heretofore inaccessible to chemists.

Table 3. Solvent Polarity Parameters for DES, Molecular Solvents, and Ionic Liquids at 293 K

DES	SP	SdP	SA	SB
reline[a]	0.877	0.721	0.677	0.359
ethaline[a]	0.816	0.975	0.546	0.379
glyceline[a]	0.851	0.878	0.583	0.436
[BMIM][BF_4][b]	0.783	0.966	0.528	0.381
[BMIM][PF_6][b]	0.728_0	0.998	0.529	0.210_4
water[c]	0.681	0.997	1.062	0.025
acetonitrile[c]	0.645	0.974	0.044	0.286
toluene[c]	0.782	0.284	0.000	0.128
DMF[c]	0.759	0.977	0.031	0.613
DMSO[c]	0.83	1.000	0.072	0.647
formamide[c]	0.814	1.006	0.549	0.414
1,2,3-propanetriol[c]	0.828	0.921	0.653	0.309
1,2-ethanediol[c]	0.777	0.91	0.717	0.534
bromobenzene	0.875	0.497	0	0.192
dibenzyl ether	0.877	0.509	0	0.33

[a]Extrapolated from temperature dependence of experimental data. [b]Reference 39. [c]Reference 33.

The SA and SB values for reline, ethaline and glyceline are comparable to those observed for ionic liquids, especially [BMIM][BF_4] ionic liquid. In particular, the SA and SB values of 0.528 and 0.381 respectively for [BMIM][BF_4] have a close correspondence with those observed for ethaline (SA = 0.546 and SB = 0.379). From the limited data available for empirical polarity of ionic liquids based on the probes proposed by Catalan and co-workers, it appears that the ionic liquids have lower SP values than the DES, but higher SdP values. Again, the paucity of data on Catalan's empirical polarity parameters of ionic liquids prevents further discussion of this aspect.

The empirical determined polarity values of the three DES were finally compared with polarity values determined by using other probe molecules, for example $E_T(30)$ parameter obtained from the solvatochromic behavior of the Reichardt's dye. The $E_T(30)$ scale is correlated with the SA, SB, and SdP parameters by the eq 2:[33]

$$E_T(30) = 23.17SA + 10.84SdP + 3.39SB + 30.5 \quad (2)$$

The $E_T(30)$ values at 298 K, thus determined, were normalized to yield the E_T^N values and are reported in Table 4 as $E_{T\,calc1}^N$. The table also lists the corresponding experimental $E_{T\,exp}^N$ values for the DES previously reported in the literature.[27] It can be observed that the values obtained from the multilinear correlation are much lower than that observed experimentally. The variation in the E_T^N values between reline, ethaline and

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Table 4. Comparison of Experimentally Reported and Calculated E_T^N Values for DES at 298 K

DES	E_T^N	$E_T^N{}_{calc1}$	$E_T^N{}_{calc2}$
reline	0.81	0.76	0.85
ethaline	0.84	0.75	0.77
glyceline	0.85	0.75	0.81

[a]Extracted from temperature dependence of previously reported experimental data in ref 27. [b]Calculated from the correlation $E_T(30)$ = 23.17 SA + 10.84 SdP+ 3.39 SB + 30.5. [c]Calculated from the correlation E_T^N = 0.74(±0.02) SA + 0.50 (±0.10) SPP − 0.12 (±0.09).

glyceline (reline< ethaline ≈ glyceline) observed experimentally, is not reflected in the calculated values either. This indicated that although the correlation may be valid for a large number of molecular solvents and ionic liquids, it may need to be modified to accommodate the data for DES.

Alternatively, one of the recent reports mentioned that the dependence of E_T^N values on empirical parameters SA and SPP could be expressed as[37]

$$E_T^N = 0.74(\pm0.02)\mathrm{SA} + 0.50(\pm0.10)\mathrm{SPP} - 0.12(\pm0.09) \quad (3)$$

However, the E_T^N values thus determined (using eq 3) and listed in Table 4 as $E_T^N{}_{calc2}$ failed to show a better correspondence with the experimental values. While the magnitude of the $E_T^N{}_{calc2}$ values was much closer to the $E_T^N{}_{exp}$ values, the fact that the second correlation also predicted reline to have the highest E_T^N values did not agree with the experimentally reported results. In order to evaluate the overall agreement or deviation between the experimental values and the calculated values of E_T^N, the temperature-dependent data were compared and a more intriguing aspect was revealed (Figure 5). Not only did the experimental and the calculated values show very poor correlation, but they also "clustered" for the three DES, indicating that the multilinear correlations employed were missing some crucial aspect related to the solvatochromism of Reichardt's dye in DES. Even if the data for reline were to be excluded from Figure 5, the comparison for ethaline and glyceline still shows a poor correlation.

Attempts to correlate the reported E_T^N values with various combinations of SA, SB, SdP, and SP parameters determined for reline, ethaline and glyceline gave the best results for the following correlation eq 4:

$$E_T^N = 0.66(\pm0.06)\mathrm{SB} + 0.01(\pm0.02)\mathrm{SdP} + 0.57(\pm0.03) \quad (4)$$

It may be possible that the indirectly determined polarity values for reline may have limited the accuracy of the attempted correlation between E_T^N and the empirical polarity parameter. The correlation analysis was repeated, this time excluding the data for reline and including only results for ethaline and glyceline, resulting in a slight improvement in the coefficients eq 5:

$$E_T^N = 0.78(\pm0.15)\mathrm{SB} + 0.08(\pm0.07)\mathrm{SdP} + 0.45(\pm0.11) \quad (5)$$

The poor correlation between E_T^N parameter and polarity values of ethaline and glyceline is consistent with the trends observed for ethaline and glyceline in Figure 5. It must be noted that the E_T^N parameter in ethaline and glyceline failed to show good correlation with SdP and SA even when the data for reline was excluded.

The results of the multiparameter linear fit are indeed very surprising and counterintuitive. On account of its structural charactersitics, Reichardt's dye is expected to be sensitive to the acidic interactions and dipolarity of the medium, with very weak sensitivity toward the basicity of the solvent. However, there have been exceptions where the stability of the dye molecule is compromised due to the acidic nature of the solvent. Concerns about the probable hydrolysis of Reichardt's dye[30] in ionic liquids have previously led to the use of an acid-resistant variant (Reichardt's dye[33]) or Nile Red for determining the polarity of ionic liquids.[27] The lack of correlation between the solvatochromic response of the betaine dye and the solvent acidity indicates the need for a detailed investigation.

Even if the dye does not undergo extensive hydrolysis, strong Coulombic interactions between the ions in the medium and dye may affect the response observed.[32] It cannot be ruled out that Reichardt's dye may undergo a strong interaction with the ions present in DES and hence, the sensitivity of the probe to the solvent properties is altered. The betaine dye is known to exhibit halochromism, i.e., strong interactions between the dye moiety and the ions present in the solvent medium. The extent of interaction with the cations has been observed to be much

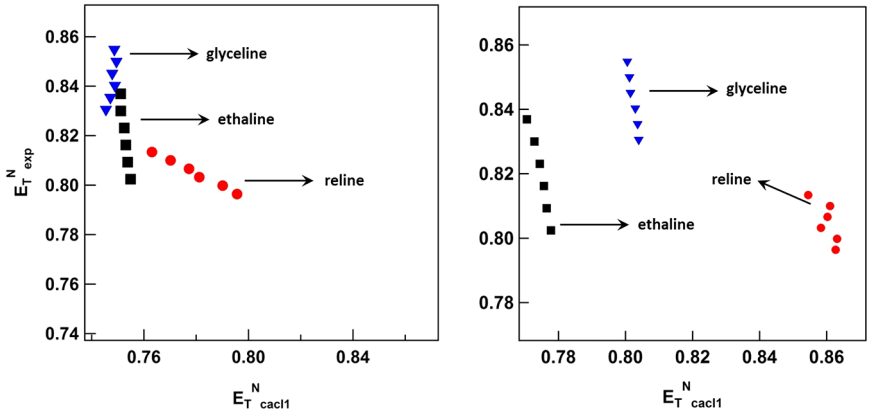


Figure 5. Correlation between experimental and calculated E_T^N values for reline, ethaline, and glyceline (note that the experimentally determined E_T^N values for all three DES decrease with increasing temperature; hence, the topmost marker for any DES indicates E_T^N value at 298 K, followed by values at steadily decreasing temperature until 323 K).

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greater, wherein the cation interacts with the EPD of the dye. The EPA moiety in betaine remains unaffected and this may reflect as greater correlation with solvent basicity. All the three DES employed in the present study consist of quaternary ammonium cation—namely, choline cation—as one of the solvent components and such cation—dye interactions cannot be completely ruled out. Very limited data is currently available on the nature of interactions between the betaine molecule and quarternary ammonium cations,[49] thus restricting the possibility of reaching any significant inferences for the intriguing results. It is probable that the compensatory effect of SA and SdP parameters in DES may negate each other and hence, are not reflected in the solvatochromic response of the betaine dye.

Finally, attempts were made to determine the SdP and SA parameters of reline using a more stable probe molecule. Earlier reports have also recommended use of alternative solvent polarity scales to determine the acidity of such solvents. For example, one report by Jessop and co-workers have recommended the use of an NMR-based scale for the experimental determination of Kamlet—Taft α vaues.[50] In the present case, the probe molecule used to determine the SA parameter (**DETZ**) is exceptionally stable to acidic environments and hence, precludes the use of alternative methods for determining the solvent acidity. The primary concern, therefore, was to explore the alternative probes for measuring solvent dipolarity in DESs. One of the potential candidates was 4-*tert*-butyl-2-(dicyanomethylene)-5-[4-(diethylamino)benzylidene]-Δ^3-thiazoline (commonly abbreviated as **Th**). Initially, the probe was introduced as a substitute for determining the Kamlet—Taft π^* or Catalan's SPP^N parameter in highly acidic solvent environments, including ionic liquids.[42,51] More recently, **Th** probe has been successfully employed in conjunction with 2-[4-(*N*, *N*-dimethylamino)benzylidene]-malononitrile (**BMN**) to measure the polarizability and dipolarity for a range of ionic liquids.[52,53] It must be noted that the approach followed by Spange and co-workers relied on multiparameter correlation of solvent polarity as a function of the probe characteristics. In a sense, this approach is similar to that adopted by Kamlet and Taft, and cannot be considered as empirical polarity values like those afforded by Catalan's probes. Accordingly, the SP and SdP parameters can be determined from the following correlations:

$$\mathrm{SP} = 0.387\bar{\nu}_{\max}(\mathbf{Th}) - 0.605\bar{\nu}_{\max}(\mathbf{BMN}) + 8.500 \quad (6)$$

$$\mathrm{SdP} = 0.556\bar{\nu}_{\max}(\mathbf{BMN}) - 0.980\bar{\nu}_{\max}(\mathbf{Th}) + 3.791 \quad (7)$$

The values of SP and SdP calculated using eq 6 and 7 respectively for the three DES at 298 K have been listed in Table 5. As an alternative approach, the solvatochromism of Th was used to estimate the SPP parameter[53] as follows:

Table 5. SP and SdP Parameters at 298 K Determined Using Th and BMN Probes for Reline, Ethaline, and Glyceline

DES	SP	SdP_{calc1}[a]	SdP_{calc2}[b]
reline	1.057	0.936	1.006
ethaline	1.138	0.856	1.079
glyceline	1.166	0.815	1.018

[a]Determined using linear correlation of SdP with ν_{Th} and ν_{BMN} eq 7 [b]Determined from the SP values (from anthracene) and SPP values (from **Th**) using eq 8

$$\mathrm{SPP}^N = 4.6542\bar{\nu}_{\max}(\mathbf{BMN}) - 0.234\bar{\nu}_{\max}(\mathbf{Th}) \quad (8)$$

The resulting SPP values alongwith the experimentally determined Catalan's SP parameter, were used to estimate the SdP parameter (note that SPP = 0.49 SP + 0.38 SdP + 0.21 which was shown to be applicable for DES in Figure 3), listed as SdP_{calc2} in Table 5.

It is immediately observed that estimates of SP based on solvatochromic behavior of **Th** and **BMN** probes are much higher than those measured by Catalan's probe system. The temperature—dependent SP values for reline and ethaline thus determined, were verified by comparing with the corresponding values obtained from refractive index measurements by using the Lorentz—Lorenz eq (Figure 6). As compared to the results in Figure 2, the SP values of reline show a poor correlation with those predicted by the Lorentz — Lorenz equation. The comparison in Figure 1 is based on the empirical SP values determined using the anthracene probe which is stable in reline and there is no reason to question their validity. It may be due to the fact that the SP values in Figure 6 are calculated using a correlation which is valid for ionic liquids but need not be valid for DESs. Similar mismatch between the π^* values based on **Th** probe and the dielectric constant reported earlier were interpreted as a result of the aromatic interactions in the solvent medium.[51] While there are no aromatic moieties in any of the three DES analyzed, the possibility of unaccounted intermolecular interactions cannot be entirely ruled out.

Since the **DMANF** probe is stable in ethaline and glyceline, it is reasonable to expect a good correlation between the SdP values determined using the different probe systems. Hence, the two sets of solvent dipolarity parameters determined on the basis of **Th** probe were compared with those based on the **DMANF** probe for mutual consistency (Figure 7). It is observed that SdP_{calc1} of glyceline shows marginally satisfactory correlation with the corresponding SdP value estimated using **DMANF**, but there is no correlation between the overall trends in the SdP values for ethaline. This is an additional evidence indicating the inconsistency between the polarity scales and emphasizing the need for "recalibration" of the polarity scales based on **Th** and **BMN** probes in DES if they are expected to be consistent with the empirical polarity values.

It may also be interesting to examine the reason for the inconsistency between the correlations applicable to ionic liquids vis-à-vis those for DES. This is a strong indication that the solvation microenvironment in the two classes of solvents may differ to a substantially greater extent than expected on the basis of our current understanding. Such a difference in the nature of solvation between DES and ionic liquids may be exploited by innovative and complementary applications of DESs as neoteric solvent media.

While more efforts are focused on the design and synthesis of new probe molecules and on the measurement of the polarity of neoteric solvents, efforts should also be made to normalize the different scales of polarity to ensure an unbiased choice of solvent which is optimum for a given application.

CONCLUSION

A detailed study of the temperature-dependent empirical polarity parameters of three deep eutectic solvents—reline, ethaline and glyceline—was undertaken. Initial observations indicated the limited stability of the **DMANF** probe molecule in reline, used for determining the SdP parameter. Extension of the correlation between the absorption behavior of **FNF** and its

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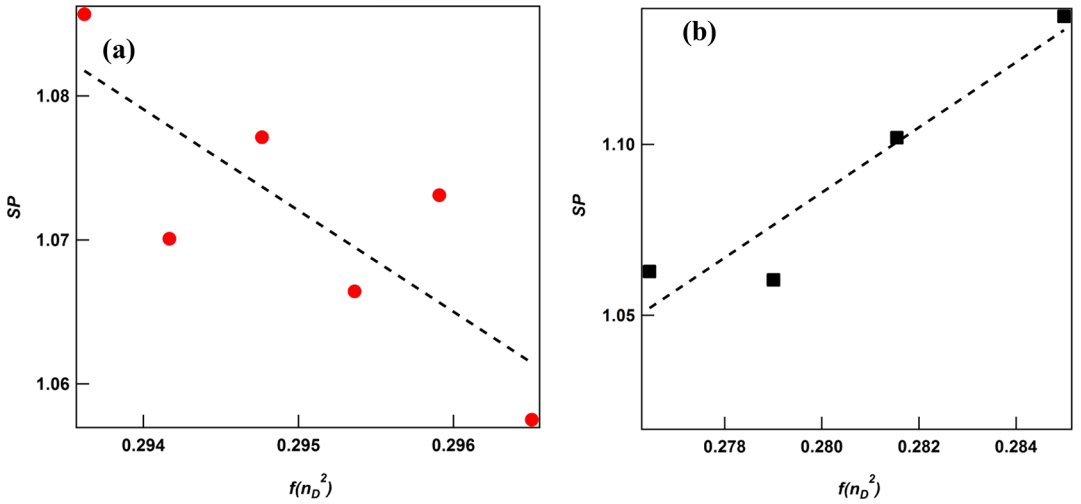


Figure 6. Correlation between temperature-dependent solvent polarizability (SP) scale determined using **Th** and **BMN** probes and the Lorentz−Lorenz function for (a) reline and (b) ethaline.

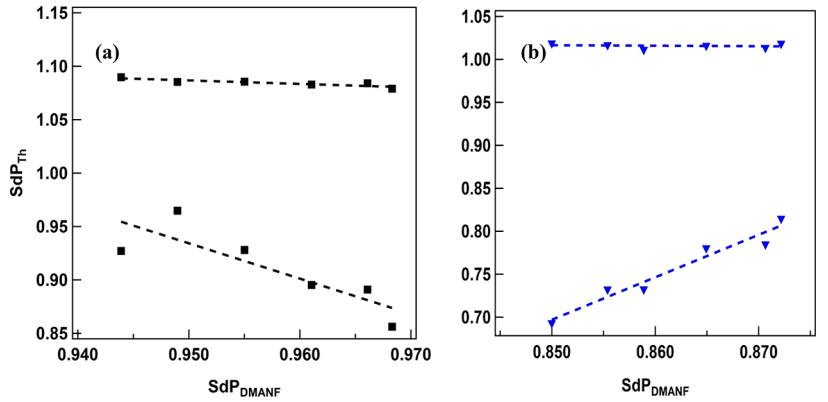


Figure 7. Correlation between temperature-dependent solvent polarizability (SP) scale determined using **Th** and **BMN** probes and the Lorentz−Lorenz function for (a) reline and (b) ethaline.

analogue **DMANF** observed in ethaline and glyceline to the data obtained for reline, facilitated the determination of SdP parameter for reline. The polarity parameters thus determined were validated by comparison with the predicted values based on the Lorentz − Lorenz equation and by comparing the predicted SPP values with those obtained from the present results.

The SA, SB, SP, and SdP parameters were compared with those obtained previously for molecular solvents and ionic liquids. It was observed that the overall polarity of DES was higher than a majority of the organic solvents. While the SdP, SA, and SB parameters of DES corresponded closely to that observed for solvents like formamide, 1,2-ethanediol, and 1,2,3-propanetriol (among other polar solvents), the SP parameter of DES was much higher. This led to the conclusion that, except for SP, the HBD component of the DES may contribute significantly to the polarity response of the probe molecule. The polarity of DES is thus unique in comparison to molecular solvents.

The temperature-dependence of the observed polarity parameters showed some interesting variations—while SP and SB parameters showed the expected decrease with increasing temperature for all three DES studied, the variation of SA and SdP parameters with temperature was not uniform. SdP of reline increased with an increase in temperature (unlike that observed for ethaline and glyceline). Although SA of reline showed a decrease in magnitude at higher temperatures, the solvent acidity of ethaline and glyceline showed an opposite trend. The positive variation in observed polarity parameters is a relatively rare property, which had been previously reported only for a few examples of ionic liquids.

Attempts to correlate the empirical polarity parameters with the E_T^N values reported in literature led to some unexpected results and interesting observations. The observations can be explained only on the basis that the solvatochromic response of betaine dye is altered in the DES environment. However, paucity of data limits the scope of discussion to a broad outline and prevents any specific attribution.

Alternative probes like **Th** and **BMN** were employed to separately determine the SdP and SP parameters in reline. But a comparison of the SP values thus determined with those determined using anthracene probe and with the Lorentz function indicated some discrepancies. Poor correlation was observed for the SdP values determined using **Th**/**BMN** and **DMANF** probe for ethaline. The most probable reason for the observed divergence may be due to the fact that, unlike Catalan's approach based on purely empirical polarity parameters, determination of polarity using Th or BMN as probes is not strictly an empirical process. The proper correlation or calibration of the spectroscopic response to the

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solvent property is critical for the accurate estimation of polarity in such cases.

Most importantly, the present work highlights the "non-classical" nature of DES, which presents chemists with challenges and opportunities. For example, it will certainly be helpful to synthesize a more stable probe to replace **DMANF** for accurate empirical determination of the SdP parameter in solvent media like reline. The challenges that need to be overcome include the development of robust polarity probes and a thorough understanding of the polarity variations with temperature, composition, etc. The exact reasons for the stabilization or destabilization of a given class of probe molecules will also shed light on the nature of interactions between the DES components and solutes—some of those may be unique to DES and not be observed for molecular solvents. Exploring the relatively unknown realm of solute−solvent interactions in DES will also pave the way for many promising applications and opportunities.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcb.7b07754.

Synthesis procedure and ^{1}H NMR characterization data of molecular probes, representative UV−vis spectrum of molecular probes, determination of solvatochromic empirical parameters, ^{1}H NMR characterization data of DES, and refractive indices of DES (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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